AHMED H. HANFY

Mechanical Engineer ||
Aerodynamic specialist

PROFILE

Doctoral researcher in the Marie Curie fellowship program with expertise in experimental aerodynamics. Proficient in Python, MATLAB, and C++; skilled in data/image analysis. Proficient in CAD modeling using Autodesk Inventor and adept with Siemens NX. Seeking to contribute expertise, pursue learning, and advance career.

CONTACT ME

Gdansk, Poland

ahmed.hady.hanfy92@gmail.com

(+48) 505288804

linkedin.com/in/engahmedhady/

github.com/EngAhmedHady/

LANGUAGES

English

Italian

Polish

Arabic

SKILLS

Ansys (Fluent)
Inventor
Siemens NX
Python
MATLAB
LabVIEW

OTHER SKILLS

Wind-tunnel operation |Test planning | Research | Manufacturing | 3D printing | Technical drawings| C++ | Fortran | OpenMPI | MATLAB GUI | PCA | Analysis | Machine learning | Image analysis | Signal processing | OOP | MS Office.

ACTIVITIES

Summer trainings [Hydro-electric stations, Diesel engines & Hydraulic maintains (MANTRAC), ASME CFD workshop].

Volunteering [researcher's night, Science Club Chairman, Torpedo robotics team: Participate in the MATE ROV competitions to perform underwater tasks].

EXPERIENCE

Doctoral Researcher (Aerodynamic specialist) [11/2020 - Present]

Institute of fluid-flow machinery polish academy of sciences, Gdansk (Poland)

Expert in transonic testing, and advanced data analysis. Actively contributed to EU research projects in collaboration with major industry partners.

Core Competencies: Wind tunnel operation, Schlieren (high-speed, multi-color), BOS, LDA, oil visualization.

Data Analysis & Software: Python tools for image/signal processing, statistics, boundary layer, isentropic flow, shock tracking.

Instrumentation & Control: LabVIEW/DAQs for feedback and camera triggering (±0.3ms trigger accuracy); flow control (suction, Reynolds).

Project Experience:

- SMS Project (Vibrating trailing edge of a morphing supercritical airfoil)
  - Performed POD analysis on transonic PIV data for noise reduction.
  - Analysed the impact of vibrating trailing edges on shock wave unsteadiness.
- TEAMAero Project (SBLI on compressor rotor profile with flow control)
  - Led test planning for fan profile surface roughness studies.
  - Coordinated the collaboration with Rolls Royce Deutschland to apply the surface texture on the suction side of the profile.
  - The study revealed a reduction in separation by 1%, and momentum thickness by 10% but increase in shock unsteadiness by 23%.
  - Implemented an inlet valve for Reynolds number control in stator testing.

SCIENTIFIC VISITS AND INTERNSHIPS

Visiting Researcher [08/2022 - 09/2022]

German Aerospace Centre (DLR), Cologne (Germany)

Within Marie Skłodowska-Curie Actions, a scientific secondment was made to DLR Transonic Cascade Wind Tunnel, successfully executed unsteady measurements, including:

- Calibration of Kulite probe and vibration sensors.
- Application analysis on high Data Acquisition rates.
- Synchronization of the speed Schlieren.

Research Internship [02/2020 - 05/2020]

Institute of fluid-flow machinery polish academy of sciences, Gdansk (Poland)

As a complementary course during master's, accomplished the following objectives:

- Proficiently extracted quantitative information from interferograms for experimental study relevant to SWBLI.
- Developed a MATLAB GUI involving FFT and phase shifting.
- Improved phase detection accuracy through the application of machine learning techniques, including DBSCAN, Linear regression, etc.

EDUCATION

Ph.D. Mechanical engineering [11/2020 - Present]

Marie Skłodowska-Curie Actions, Innovative Training Networks HORIZON 2020

Institute of fluid-flow machinery polish academy of sciences, Poland

(Est. Graduation, late 2024)

M.Sc. Applied Mathematics and Mathematical Engineering

InterMaths Joint MSc Programme [09/2018 - 05/2020]

- Ivan Franko National University of Lviv (IFNUL), Ukraine [2019 - 2020]
- University of L'Aquila (UAQ), Italy [2018 - 2019]

B.Sc. Mechanical Engineering [09/2011 - 07/2016]

Alexandria University (AU), Egypt

RECENT PUBLICATIONS

- Hanfy, A. H., Flaszynski, P., Doerffer, P., & Kaczynski, P. (2026). Enhanced line-scanning method for shock wave tracking in high-speed schlieren imaging. Aerospace Science and Technology, 111022.
- Hanfy, A.H., Flaszyński, P., Kaczyński, P., Doerffer, P. (2025). Surface Roughness Effect on Shock Boundary Layer Interaction on Compressor Rotor Profile. In: Flaszynski, P. et al (eds). Towards Effective Flow Control and Mitigation of Shock Effects in Aeronautical Applications. vol 201. Springer, (P. 325–344).
- Nel, P. L., Janke, C., Vasilopoulos, I., Swoboda, M., Schreyer, A.-M., Hady, A., & Flaszyński, P. (2024). Effect of transition on self-sustained shock oscillations in highly loaded transonic rotors. AIAA Journal, 0(0):1–13.

SELECTED PROJECTS

- Shock Tracking Library (OpenCV, Python)
- Finite Fringe Analysis for Optical Measurement of Compressible Fluid Flow Parameters (MATLAB GUI application - MSc. Thesis)
- Parallel implementation of Poisson's equation (OpenMPI, Fortran and C++)

Full CV